```
mkdir app/Services
touch app/Services/CalculatorService.php
```

Edit `app/Services/CalculatorService.php`:

```
<?php

namespace App\Services;

use App\DTOs\CalculatorDTO;

class CalculatorService
{
    public function calculate(CalculatorDTO $dto): float
    {
        switch ($dto->operation) {
            case 'add':
                return $dto->num1 + $dto->num2;
            case 'subtract':
                return $dto->num1 - $dto->num2;
            case 'multiply':
                return $dto->num1 * $dto->num2;
            case 'divide':
                if ($dto->num2 == 0) {
                    throw new \Exception("Cannot divide by zero");
                }
                return $dto->num1 / $dto->num2;
            default:
                throw new \Exception("Invalid operation");
        }
    }
}
```

14.11.4.5 Step 5: Create the Calculator API Controller

Run the following command to create a **CalculatorController**:

```
php artisan make:controller CalculatorController
```

Now, edit `app/Http/Controllers/CalculatorController.php`:

```
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;
use App\DTOs\CalculatorDTO;
use App\Services\CalculatorService;
use Illuminate\Http\JsonResponse;

class CalculatorController extends Controller
{
    private $calculatorService;

    public function __construct(CalculatorService $calculatorService)
    {
        $this->calculatorService = $calculatorService;
    }
```